hw 1

Ian Chi

January 14, 2026

1. (a) the sum over all x and y should add up to 1.

$$\sum_{x\in\{0,1,2\}}\sum_{y\in\{0,1\}} c(x+y) = 1 \tag{1}$$

this implies $c = \left(\sum_{x\in\{0,1,2\}}\sum_{y\in\{0,1\}}(x+y)\right)^{-1}$ $c = 1/9$

(b)

$$P(x) = \sum_y c(x+y) = c(x+1) \tag{2}$$

$$P(y) = \sum_x c(x+y) = c(y+3) \tag{3}$$

(c)

$$P(x|y=1) = \sum_x c_2(x+1) \tag{4}$$

where $c_2 = 1/6$

(d) no. the distribution for x changes based on whether y is 0 or 1.

2. (a)

$$\begin{gathered} P_{z_i}(1) = p \\ P_{z_i}(0) = 1-p \end{gathered} \tag{5}$$

(b) the pmf just follows a binomial distribution with $p = p$

$$P(x) = \binom{n}{x} p^x (1-p)^{n-x} \tag{6}$$

$$P(x \geq k) = \sum_k^n \binom{n}{x} p^x (1-p)^{n-x} \tag{7}$$

(c)

$$P(Z_1 = 1|S_n \geq 1) = \frac{P(S_n \geq 1|Z_1 = 1)P(Z_1)}{P(S_n \geq 1)} = \frac{1 \cdot p}{\sum_k^n \binom{n}{x} p^x (1-p)^{n-x}} \tag{8}$$

the probability changes since we know one at least is true, each one is more likely to be working.

(d)

$$P(T > m|S_n \geq 1) = \frac{1}{(1-p)^T} P(Z_1 = 1|S_{n-m} \geq 1) \tag{9}$$

this question is the same as the original problem except the first m failed. the probability that the first m failed is $\frac{1}{(1-p)^T}$ thus we multiply them together and they are independent since no packet affects the outcome of the next packet.

3. (a) the distribution is an exponential distribution but discrete.

$$P(T = k) = (1-p)^{k-1} p \tag{10}$$

(b)

$$1 - \sum_{i=0}^{k} (1-p)^{i-1} p = (1-p)^k \tag{11}$$

(c) apply

$$\sum k r^{k-1} = \frac{1}{(1-r)^2} \tag{12}$$

for r between 0 and 1. since 1-p is between 0 and 1,

$$\mathbb{E}(T) = \sum k P(T = k) = \frac{1}{p} \tag{13}$$

(d)

$$\begin{aligned} P(T > m + n|T > m) &= \frac{P(T > m|T > m + n)P(T > m + n)}{P(T > m)} \\ &= \frac{1 \cdot (1-p)^{m+n}}{(1-p)^m} = (1-p)^n \end{aligned} \tag{14}$$

it means we sort of can restart our algorithm at the beginning if something fails before.

4. (a) the probability is $\frac{1}{100}$.

   (b) the probability is concentrated into one door and thus the probability is $\frac{99}{100}$.

(c) staying always results in $\frac{1}{100}$. swapping gives probability

$$\frac{99}{100}\frac{1}{99-k} \tag{15}$$

where k is the number of doors opened. the probability is concentrated into $99-k$ doors so the probability is distributed $\frac{1}{99-k}$ ways.

(d) yes. the probability in the equation above is greater than $\frac{1}{100}$ for all $k \geq 1$

5. (a) since each person removes a month from the possibilities of the next persons birthday we see the probability is

$$\frac{12!}{(12-n)!(12)^n} \tag{16}$$

for $n \leq 12$. then 0 for $n > 12$.

(b) the probability is

$$1 - \frac{12!}{(12-n)!(12)^n} \tag{17}$$

. each person removes a 1/12 probability that the next person has of colliding.

(c) it takes 5 people. p(same birth month with 5 people) = .61806

(d) the probability 3 birthdays collide is $\frac{1}{12}^2$. there should be approx $\binom{n}{3}$ ways fo this to happen, thus

$$P_3(n) = \frac{\binom{n}{3}}{144} \tag{18}$$